

SL2DT-08: Automation & GitOps

Series: SL2DT | **Notebook:** 8 of 10 | **Created:** April 2026 | **Last Updated:** 04/21/2026

Overview

Goal of this step: automate the promotion path for dashboards, monitors, workflows, IAM, OpenPipeline, and bucket configs. Migration is only sustainable if post-cutover changes go through the same CI/CD discipline — not via UI clickops that drift out of source control.

Automation applies both to the migration itself (programmatically import 8,000 dashboards faster than clicking) and to ongoing operations (every config change is a PR).

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Prerequisites

Requirement	Details
Audience	Platform engineering, DevOps, SRE
Tools	Git, Terraform ≥1.6, Monaco ≥2.x, Python 3, cloud CLIs (AWS/GCP/Azure)
Dynatrace access	Platform Token with broad write scopes; separate service account for CI
Prior reading	AUTOM-01 (Configuration automation overview), K8S-07 (GitOps for Dynatrace), <code>Dynatrace-NewRelic</code> for migration-tool patterns

1. What You'll Produce

Artifact	Purpose
<code>terraform/</code>	All IaC for buckets, IAM, OpenPipeline, Workflows
<code>monaco/</code>	Bulk configs (dashboards, monitors, settings)
<code>.github/workflows/</code> (or equivalent CI)	Pipelines for validate → plan → apply
<code>scripts/extract-and-import/</code>	Sumo → DT migration pipeline scripts
<code>runbooks/rollback.md</code>	How to revert a bad config change

2. Tool Selection — Terraform vs Monaco vs the Future Migration Tool

Tool	Use for	Avoid for
Terraform (Dynatrace provider)	Buckets, IAM groups/policies, Settings 2.0 schemas, some Workflows	Bulk dashboards (slow), complex dashboard JSON

Tool	Use for	Avoid for
Monaco	Dashboards, monitors, settings in bulk; promotion across environments	IAM (use Terraform)
Dynatrace-SumoLogic migration tool (future — see AGENT-TASKS.md)	Sumo extract + translate + import	Ongoing change management
Direct API (scripts)	One-offs, custom orchestration, Document API for notebooks	Anything that needs state tracking
Dynatrace UI	Investigation, exploration — never for config changes post-cutover	Migration execution

Principles

- **One tool per concern.** Don't manage the same asset in both Terraform and Monaco.
- **State reviewable.** Every config change goes through PR with `terraform plan` or `monaco deploy --dry-run` output.
- **Idempotency.** Every pipeline must be safe to re-run.

3. Terraform — Bucket, IAM, Settings 2.0 Config

Provider Setup

```
terraform {
  required_providers {
    dynatrace = {
      source = "dynatrace-oss/dynatrace"
      version = "~> 1.60"
    }
  }
}

provider "dynatrace" {
  dt_env_url      = var.dt_env_url
  dt_platform_token = var.dt_platform_token # Bearer token
}
```

Organizing Modules

```
terraform/
├─ modules/
│   ├─ bucket/           # Grail buckets
│   ├─ iam-group-policy/ # Group + policy + binding
│   ├─ openpipeline/     # Pipelines + processors
│   └─ workflow/         # Workflow configs
├─ envs/
│   ├─ prod/
│   │   └─ main.tf
│   ├─ preprod/
│   └─ dev/
└─ variables.tf
```

Example — Bucket Module Call

```
module "bucket_prod_logs" {
  source      = "../../modules/bucket"
  name        = "custom_logs_prod"
  display_name = "Production Logs"
  retention    = 90
}

module "iam_prod_readers" {
  source      = "../../modules/iam-group-policy"
  group_name   = "g_prod_readers"
  description  = "Read-only prod logs"
  policy_statements = [
    "ALLOW storage:logs:read WHERE storage:bucket-name = \"custom_logs_prod\";"
  ]
}
```

State Management

Use a remote backend (S3 + DynamoDB lock, or GCS, or Terraform Cloud):

```
terraform {
  backend "s3" {
    bucket      = "tfstate-dynatrace-migration"
    key         = "prod/dynatrace.tfstate"
    region      = "us-east-1"
    dynamodb_table = "tfstate-lock"
  }
}
```

Never commit `.tfstate` files to git.

4. Monaco — Bulk Config Promotion

Monaco is the right tool for dashboards/monitors in bulk. It excels at multi-tenant promotion (dev → preprod → prod).

Project Structure

```
monaco/
├─ environments.yaml      # Tenant list + auth
├─ projects/
│   └─ dashboards/
│       └─ config.yaml
│       └─ definitions/  # One JSON/YAML per dashboard
│   └─ monitors/
│   └─ openpipeline/
```

environments.yaml (non-secret)

```
environmentGroups:
- name: production
  environments:
  - name: prod
    url:
      type: value
      value: https://&lt;env-id&gt;.apps.dynatrace.com
    auth:
      token:
        type: environment
        name: DT_PROD_TOKEN
- name: preprod
  environments:
  - name: preprod
  ...
```

Deploying Dashboards

```
monaco deploy -e environments.yaml projects/dashboards --environment-group
production --dry-run
monaco deploy -e environments.yaml projects/dashboards --environment-group
production
```

Template Variables

Monaco supports per-environment templating:

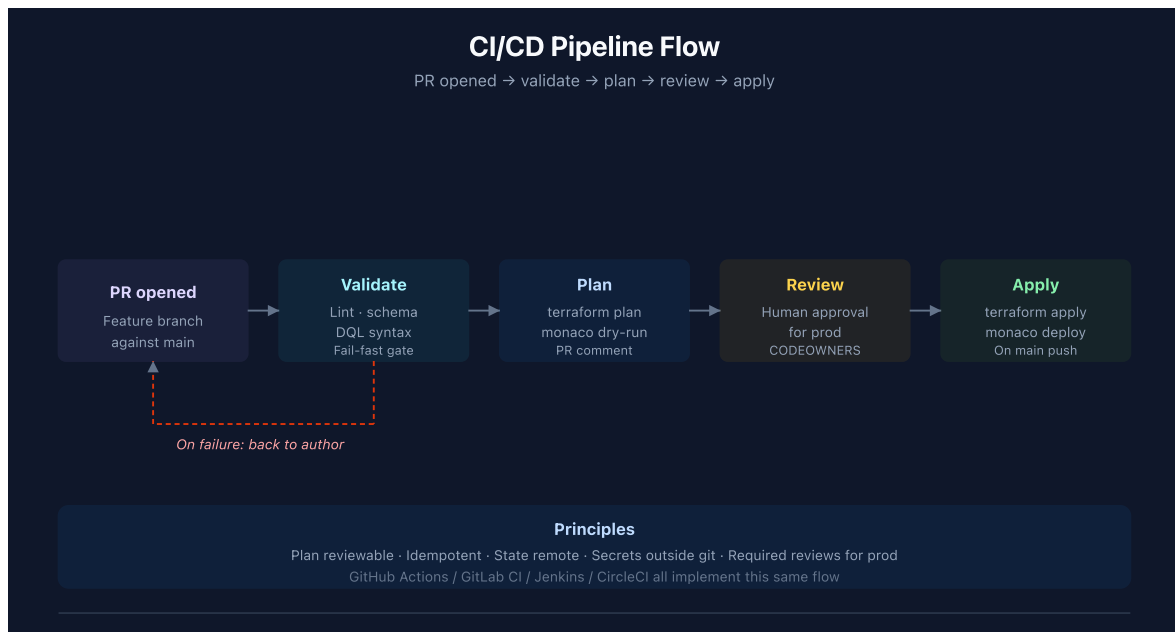
```
configs:
- id: dashboard_payments_api
  type: dashboard
  template: templates/payments-api.json
  parameters:
    tenant_name: "{{ .name }}"
    prod_bucket: "custom_logs_prod"
    env_label: "{{ .environmentName }}"
```

Migration Use Case — Bulk Dashboard Import

If you have 2,000 translated dashboards, iterating via Terraform's `dynatrace_document` is slow (one API call per resource, sequential apply). Monaco batches and deploys in parallel — typical 5–10x faster for bulk work.

5. CI/CD Pipeline Patterns

Typical Pipeline Stages



GitHub Actions Example

```
name: dynatrace-config
on:
  pull_request:
    branches: [main]
  push:
    branches: [main]

jobs:
  validate:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - run: python3 scripts/validate-dql.py terraform/
      - run: monaco lint monaco/
      - uses: hashicorp/setup-terraform@v3
      - run: |
          cd terraform/envs/prod
          terraform init -backend=false
          terraform validate

  plan:
    if: github.event_name == 'pull_request'
    needs: validate
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: hashicorp/setup-terraform@v3
      - run: |
          cd terraform/envs/prod
          terraform init
          terraform plan -no-color > plan.txt
      - uses: actions/github-script@v7
        with:
          script: |
            const fs = require('fs');
            const plan =
fs.readFileSync('terraform/envs/prod/plan.txt', 'utf8');
            github.rest.issues.createComment({
              issue_number: context.issue.number,
              owner: context.repo.owner, repo: context.repo.repo,
              body: ``\n' + plan + '\n``
            });

  apply:
    if: github.event_name == 'push'
```



```
needs: validate
runs-on: ubuntu-latest
environment: production
steps:
  - uses: actions/checkout@v4
  - uses: hashicorp/setup-terraform@v3
  - run: |
      cd terraform/envs/prod
      terraform init
      terraform apply -auto-approve
  - run: monaco deploy -e environments.yaml monaco/projects --
environment-group production
```

Secrets Management

Platform Tokens and API keys live in:

- **GitHub Actions:** repository secrets
- **GitLab CI:** protected variables
- **Jenkins:** Credentials plugin
- **CircleCI:** contexts

Never commit tokens. Every `*.env` file goes in `.gitignore`.

6. Automating the Sumo Extract → DT Import Flow

For bulk migration work (Waves 3 and 4), scripts automate the extract + translate + import loop. The eventual `Dynatrace-SumoLogic` migration tool (see `docs/AGENT-TASKS.md`) is the destination; until built, scripts.

Minimal Python Pipeline

```
#!/usr/bin/env python3
"""Extract Sumo dashboards, translate, import to Dynatrace."""
import os, json, requests, base64
from pathlib import Path

SUMO_AUTH = base64.b64encode(
    f'{os.environ['SUMO_ACCESS_ID']}:
{os.environ['SUMO_ACCESS_KEY']}'.encode()
).decode()
DT_TOKEN = os.environ['DT_TOKEN']    # dt0s16.XXX
DT_URL    = os.environ['DT_TENANT']
SUMO_URL  = f'https://api.{os.environ['SUMO_REGION']}.sumologic.com'

def sumo_get(path):
    r = requests.get(f'{SUMO_URL}{path}',
                     headers={"Authorization": f"Basic {SUMO_AUTH}"})
    r.raise_for_status()
    return r.json()

def dt_post(path, body):
    r = requests.post(f'{DT_URL}{path}', json=body,
                     headers={"Authorization": f"Bearer {DT_TOKEN}"})
    r.raise_for_status()
    return r.json()

def translate_dashboard(sumo_dashboard):
    # Uses sumoql-to-dql skill mapping tables
    # Returns DT notebook/dashboard JSON
    ...

def main():
    out = Path("./translated")
    out.mkdir(exist_ok=True)
    for dash in sumo_get("/api/v2/dashboards")["dashboards"]:
        dt_config =
translate_dashboard(sumo_get(f"/api/v2/dashboards/{dash['id']}"))
        (out / f'{dash['id']}.json").write_text(json.dumps(dt_config,
indent=2))
        # Dry-run by default; --apply flag pushes to DT
        # dt_post("/platform/document/v1/documents", dt_config)

main()
```

Preflight Check (before bulk import)

Adopt the **PreflightCheck** pattern from the future migration tool (see `docs/AGENT-TASKS.md §3`) for script-level migrations:

```
@dataclass
class PreflightCheck:
    name: str
    status: str          # pass / warn / fail
    scopes_min: list[str]
    scopes_recommended: list[str]
    diagnosis: str
    remediation: str

def preflight():
    checks = []
    # Check token prefix determines auth scheme
    prefix = DT_TOKEN[:7]
    if prefix.startswith("dt0s16"):
        auth_scheme = "Bearer"
    elif prefix.startswith("dt0c01"):
        auth_scheme = "Api-Token"
    else:
        checks.append(PreflightCheck(
            "token_prefix", "fail",
            [], [],
            f"Unrecognized token prefix: {prefix}",
            "Generate a Platform Token (prefix dt0s16) or Classic API Token (prefix dt0c01)"
        ))
    # ... more checks
    return checks
```

Auth Scheme by Token Prefix

Critical pattern — every HTTP call must pick the right header:

Token Prefix	Scheme
dt0s16.	Authorization: Bearer <token>
dt0s01.	Authorization: Bearer <token> (OAuth)
dt0c01.	Authorization: Api-Token <token>

Wrong scheme → `401 Unsupported authorization scheme` , even with every scope.

7. Ansible / CloudFormation / Cloud-Specific Handoffs

Many customers run existing automation for infrastructure. Integrate rather than replace.

Ansible (host-based automation)

OneAgent installation via Ansible playbook:

```
- name: Install Dynatrace OneAgent
  hosts: all
  become: true
  tasks:
    - name: Download OneAgent installer
      get_url:
        url: "{{ dt_installer_url }}"
        dest: /tmp/oneagent.sh
        headers:
          Authorization: "Api-Token {{ dt_pass_token }}"
    - name: Install OneAgent
      shell: /bin/sh /tmp/oneagent.sh --set-host-group="{{ dt_host_group }}"
      args: { creates:
        /opt/dynatrace/oneagent/agent/lib64/liboneagentloader.so }
```

CloudFormation / AWS

Use the Dynatrace AWS Clouds app (native integration) rather than custom lambdas. Clouds app push path is settled practice — no ActiveGate needed for most AWS metrics.

Terraform for AWS role creation:

```
resource "aws_iam_role" "dynatrace_monitoring" {
  name = "DynatraceMonitoringRole"
  assume_role_policy = jsonencode({
    Version = "2012-10-17"
    Statement = [{
      Effect    = "Allow"
      Principal = { AWS = "arn:aws:iam::509560245411:role/dynatrace-
monitoring-role" }
      Action    = "sts:AssumeRole"
      Condition = { StringEquals = { "sts:ExternalId" = var.external_id } }
    }]
  })
}
```

Azure / GCP

Native cloud integrations via Dynatrace tenant UI or API. Avoid custom exporters unless absolutely needed.

Decision Matrix

Source Platform	Ingest Path
On-prem Linux/Windows hosts	OneAgent via Ansible
AWS EC2 + services	OneAgent (from host) + AWS Clouds app (for services)
AWS Lambda	Dynatrace Lambda extension
Azure VM + services	OneAgent + Azure integration
GCP GCE + services	OneAgent + GCP integration
Kubernetes	DynaKube operator
CloudWatch / EventBridge	AWS Clouds app

8. Post-Migration: Change Management

Once Sumo is decommissioned (SL2DT-09), every Dynatrace config change must go through the same pipeline.

The New Baseline

- All monitors, dashboards, buckets, IAM, OpenPipeline configs in git
- Every change is a PR
- No UI clickops for production changes (UI for exploration only)

Enforcement

- CI blocks any PR with invalid DQL, schema errors, or bucket misconfig
- Required reviews: 1 engineer + 1 platform-team approver for prod
- Settings 2.0 schema drift detection: weekly automated diff between tenant state and git state

Drift Detection Workflow

```
# .github/workflows/drift-detect.yml
name: drift-detect
on:
  schedule:
    - cron: "0 6 * * 1" # Weekly Monday
jobs:
  drift:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - run: terraform plan --detailed-exitcode
      - run: monaco deploy --dry-run --report drift-report.md
      - if: failure()
        uses: actions/github-script@v7
        with:
          script: |
            github.rest.issues.create({
              owner: context.repo.owner, repo: context.repo.repo,
              title: 'Dynatrace config drift detected',
              body: 'See attached diff'
            });
```

9. Step Exit Criteria

- [] All buckets + IAM + OpenPipeline managed via Terraform
- [] Dashboards + monitors managed via Monaco
- [] CI pipeline runs on every PR: validate → plan → apply
- [] Secrets managed outside git (CI vault, not `.env` in repo)
- [] Preflight-check pattern used in all migration scripts
- [] Ansible / CloudFormation integration documented per source platform
- [] Drift detection scheduled + alerting on failure
- [] Rollback runbook in `runbooks/rollback.md`

Next step: SL2DT-09 — Cutover, Validation & Decommission (parallel run, validation gates, Sumo teardown).

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